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PAPER_955: BCS Phonon Resonance at ω_{SCm}

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** et_{phonon_resonance}.py §6 (BCSPhononResonance) **Calculator:** BCSPhononResonanceCalc (CP4 #539) **CVW:** v2.0.0 compliant

Abstract

We analyze the BCS superconducting gap at the SCm phonon resonance frequency $\omega_{\text{extSCm}} = 2\pi \times 1.25$ THz. The gap equation $\Delta = (\hbar\omega_{\text{SCm}}/2) \tanh(\Delta/2k_B T) \cdot S_{26} \cdot (F_{UBi}/F_U)$ is solved self-consistently at the phonon resonance, revealing how the SCm vacuum phonon mode mediates Cooper pairing.

1. BCS Gap at Phonon Resonance

$$\Delta_{\text{extres}} = \frac{\hbar\omega_{\text{SCm}}}{2} \cdot \tanh\left(\frac{\Delta}{2k_B T}\right) \cdot S_{26} \cdot \frac{F_{UBi}}{F_U}$$

2. Iterative Solution

Starting from $\Delta_0 = \hbar\omega_{\text{SCm}}/2$, iterate:

$$\Delta_{n+1} = \text{rhs}(\Delta_n, T)$$

until $|\Delta_{n+1} - \Delta_n| < \epsilon$.

3. Phonon Enhancement Factor

The resonance Q-factor at the phonon peak:

$$Q_{\text{res}} = \frac{\omega_{\text{extSCm}} \cdot \sqrt{\Delta}}{k_B T}$$

4. Source Data

- **File:** et_{phonon_resonance}.py §6
 - **Session:** 214
 - **CP4 Class:** BCSPhononResonanceCalc (#539)
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References

1. Bardeen, Cooper, Schrieffer — Theory of Superconductivity (1957)
2. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
3. PAPER_949 — BCS Gap Equation
4. PAPER_950 — BCS Critical Temperature
5. PAPER_951 — Cooper Pair Phonon Coupling
6. PAPER_956 — Spectral Ladder Phonon Mapping

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation solved at resonance
PAPER_951	Coupling strength at phonon peak
PAPER_956	26-level phonon mapping uses this Q
PAPER_952	Spectral ladder energies

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper's domain.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)

Sector	Domain	Late-Corpus Result
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component rho (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^3} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

This sum converges absolutely for $||[\text{SSq}]|| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day ⁻¹	Magnetar spin-down
$[\text{SSq}]$	—	0.57	BH dynamics
ω_{textSCm}	—	$2\pi \times 1.25$ THz	Resonance frequency
F_{UBi}/F_U	—	0.6	Buoyancy ratio

Observable	UQFF Prediction	Status
$\Delta_{textres}$ at $\omega_{textSCm}$	Self-consistent via fixed-point	Validated
Phonon Q-factor	$Q = \omega_{textSCm} \sqrt{\Delta} / (k_B T)$	Derived

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Phonon Resonance (BCS at $\omega_{textSCm}$)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{res}} = \mathcal{L}_{\text{gap}}|_{\omega=\omega_{textSCm}} + \frac{1}{2} \hbar \omega_{\text{SCm}} \coth! \left(\frac{\hbar \omega_{\text{SCm}}}{2k_B T} \right)$$

§A.3 Euler-Lagrange Equation of Motion

$$\Delta_{textres} = \frac{\hbar \omega_{\text{SCm}}}{2} \tanh! \left(\frac{\Delta}{2k_B T} \right) S_{26} \frac{F_{UBi}}{F_U}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ phonon peak $\rightarrow$ BCS gap at resonance $\rightarrow$ maximal Cooper pairing

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

On-resonance VDS peak: $\rho_{textSCm} \cdot S_{26}$ (maximal phonon occupation).

§B.2 DVP

Resonance prime $p = 2$ (Cooper pair at peak coupling).

§B.3 BSH

Q_{res} determines the BSH bandwidth at the resonance shell.

§B.4 Production-Scale Consistency

Metric	Value	Status
$\omega_{textSCm}$	$2\pi \times 1.25$ THz	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

12 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
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5. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179
6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
7. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1